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SMPR: a structure-enhanced multimodal drug–disease prediction model for drug repositioning and cold start

Xin Dong[1,2,5], Rui Miao[3], Suyan Zhang[4], Shuaibing Jia[5], Leifeng Zhang[5], Yong Liang[6*], Jianhua Zhang[2,5*] and Yi Zhun Zhu[2*]

Abstract

Repositioning drug–disease relationships has always been a hot field of research. However, actual cases of biologically validated drug relocation remain very limited, and existing models have not yet fully utilized the structural information of the drug. Furthermore, most repositioning models are only used to complete the relationship matrix, and their practicality is poor when dealing with drug cold start problems. This paper proposes a structure-enhanced multimodal relationship prediction model (SMRP). SMPR is based on the SMILE structure of the drug, uses the MOL2VEC method to generate drug-embedded representations, and learns disease-embedded representations through heterogeneous network graph neural networks. Ultimately, a drug–disease relationship matrix is constructed. In addition, to reduce the difficulty of use, SMPR also provides a cold start interface on the basis of structural similarity based on repositioning results to predict drug-related diseases simply and quickly. The repositioning ability and cold start capability of the model were verified from multiple perspectives. While the AUC and ACUPR scores of repositioning reached 99% and 61%, respectively, the AUC of the cold start method was 80%. In particular, the cold start recall indicator can reach more than 70%, which means that the SMPR is more sensitive to positive samples. Finally, case analysis was used to verify the practical value of the model, and visual analysis directly demonstrated the improvement in the structure of the model. For ease of use, we also provide local deployment of the model and packaged it into an executable program.

Scientific contribution

The SMPR model is a structure-enhanced multimodal learning model that focuses on the reference value of similar structures in practical docking and adds a drug embedding module. Considering that the predictive ability of the model is limited by the dataset, the model provides a cold start interface for the effective prediction of drugs that are not in the dataset. To further facilitate the use by pharmacology workers without programming knowledge and enhance its practical application value, the model was also encapsulated into a local executable program.

Keywords Multimodal data, Drug SMILE structure, Repositioning, Cold start

*Correspondence:
Yong Liang
yongliangresearch@gmail.com
Jianhua Zhang
petermails@zzu.edu.cn
Yi Zhun Zhu
yzzhu@must.edu.mo
Full list of author information is available at the end of the article

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Introduction

The development of new drugs is an enormous challenge in terms of time and cost [11]. Using computer models to describe the complex relationships among drugs, diseases, and targets can accelerate the research process in terms of effectiveness, toxicity, and other aspects. Especially after the COVID-19 pandemic in 2019, methods such as drug repositioning to accelerate drug development have become even more important [19, 21]. Entecavir, penciclovir, ganciclovir, indinavir, and other drugs have been reported to have potential effects on COVID-19 [40]. However, drugs that have been practically applied through drug repositioning are still limited, and many drugs are still in clinical trials or are abandoned at stage III of clinical trials [45]. Therefore, supplementing information to further improve the accuracy of drug repositioning models remains a direction worthy of attention.

Multimodal data, which add additional prior knowledge to the prediction of drug–disease relationships, provide a new perspective for repositioning drugs from different data sources [23, 56]. The deepDR proposed by Zeng et al. integrates seven relational networks, including drug, disease, side effects, and target networks, to predict potential associations between drugs and diseases [59]. The NCH-DDA model uses node aggregation and neighborhood fusion to further enhance information exchange between different modalities [60]. In particular, the REDDA model proposed by Gu et al. uses five entities, including drugs, proteins, genes, pathways, and diseases, to construct heterogeneous networks, greatly enriching the knowledge that the model can learn [13].

However, most of the current multimodal data-based repositioning models still lack sufficient consideration of the importance of drug structure, where compound structures are only used to construct network relationships between drugs. For the same protein active pocket, it is particularly important to determine whether the drug can enter smoothly and has a group that can form a stable chemical bond with the residues therein [3, 42]. In addition, the purpose of the repositioning model task is to complete the drug–disease relationship. The existing drug repositioning models are limited by the size of the training dataset and lack practical application extensions, which severely limits the model's generalizability [29]. In particular, with the development of AI-assisted drug design, the drug development process has greatly accelerated, leading to the emergence of many new drugs. This has resulted in many excellent models being powerless in analysis of new drugs. The prediction of new drugs with limited prior knowledge is called a cold start (which does not exist in the model dataset), where the prediction of the relationship between unknown drugs and diseases is called a drug cold start [14].

To address the issues of repositioning mentioned above, we propose the SMPR model. The model includes a drug embedding module and a disease embedding module, which is an improvement on the existing model. The drug embedding module is inspired by natural language processing (NLP), treating the SMILE structure of compounds as a natural language and generating unique embedding representations for each compound using the MOL2VEC model. The disease embedding module utilizes heterogeneous networks to integrate multimodal information and jointly construct vector representations of diseases. Finally, the model reconstructs the drug–disease relationship on the basis of the features obtained from the two modules. To solve the cold start problem, we provide a simple cold drug start method based on the model repositioning learning results. For a drug that is not in the dataset, the drug embedding module is used to quickly obtain its representation, and on the basis of the structural similarity weighting and prior knowledge of some existing diseases, the most likely disease to act is recommended.

Materials and methods

Datasets

We collected two datasets for training and testing the model reposition and cold start modules (Table 1). Dataset A (DA) was derived from the Fdataset [12], Cdataset [32], and KEGG [22] databases and contains 894 drugs, 454 diseases, and 2704 confirmed drug–disease associations. More diverse knowledge of drugs, proteins, genes, pathways, and drug connectivity was sourced from the DrugBank [24], CTD [9], STRING [52], and UniProt [53] databases. As shown in Gu et al. [13], data preprocessing was carried out. In addition to known drug–protein, disease–gene, and drug–disease relationships, drug–drug and disease–disease matrices were constructed on the basis of drug extended

Table 1 Introduction of Dataset A, CS_train and CS_test

	Dataset A	$Cold_start_{train}$	$Cold_start_{test}$
Drug SMILE structures	894	805	89
Drug–Drug R_{rr}	799,236	627,653	0
Drug–Protein R_{rp}	4397	3956	0
Protein–Gene R_{pg}	18,545	18,545	0
Gene–Pathway R_{gw}	25,995	25,995	0
Pathway–Disease R_{wd}	19,530	19,530	0
Disease–Disease R_{dd}	42,500	42,046	0
Drug–Disease R_{rd}	2704	2425	279

connectivity fingerprint (ECFP) similarity and disease medical subject heading (MeSH) similarity.

ECFP similarity, which is based on the Tanimot score, is a widely recognized method for relocation tasks [6, 20, 28, 37]. SMILE is represented in the form of ECFPs combined with Tanimot correlation functions provided by the RDKit library to obtain drug-to-drug similarity. Next, the disease information data, that is, MeSH, were sourced from the National Library of Medicine, which hierarchically correlates diseases in the form of trees. In previous studies, the relationships between diseases were often constructed in the form of directed acyclic graphs (DAGs), where diseases were hierarchically associated with unique diseases [16, 28, 43]. When evaluating the similarity between two diseases, the similarity score is obtained by comparing the overlapping connection relationship between the two.

To validate the cold start capability of the model, we divided Dataset A into two parts: $Cold_start_{train}$ (CS_train) and $Cold_start_{test}$ (CS_test). DA was divided at a ratio of 9:1, where CS_train contained 805 drugs and CS_test contained 89 drugs. When the cold start was validated later, the model was retrained on CS_train, and CS_test was treated as a new drug that did not appear in the training data.

In addition, to verify the stability of the model, we collected the dataset used by Zhao et al. [61] to construct Dataset C (DC), in which the data only included 579 drugs and 274 diseases, as shown in Table 2. We supplemented the structural information corresponding to the drugs.

Table 2 Introduction of Dataset C

	Dataset C
Drug SMILE structures	579
Drug-Protein R_{rp}	3219
Disease-Protein R_{dp}	43,131
Drug-Disease R_{rd}	1897

Model framework

The SMRP model, as shown in Fig. 1, mainly includes a disease embedding module and a drug embedding module. In this chapter, we provide a detailed introduction to each module.

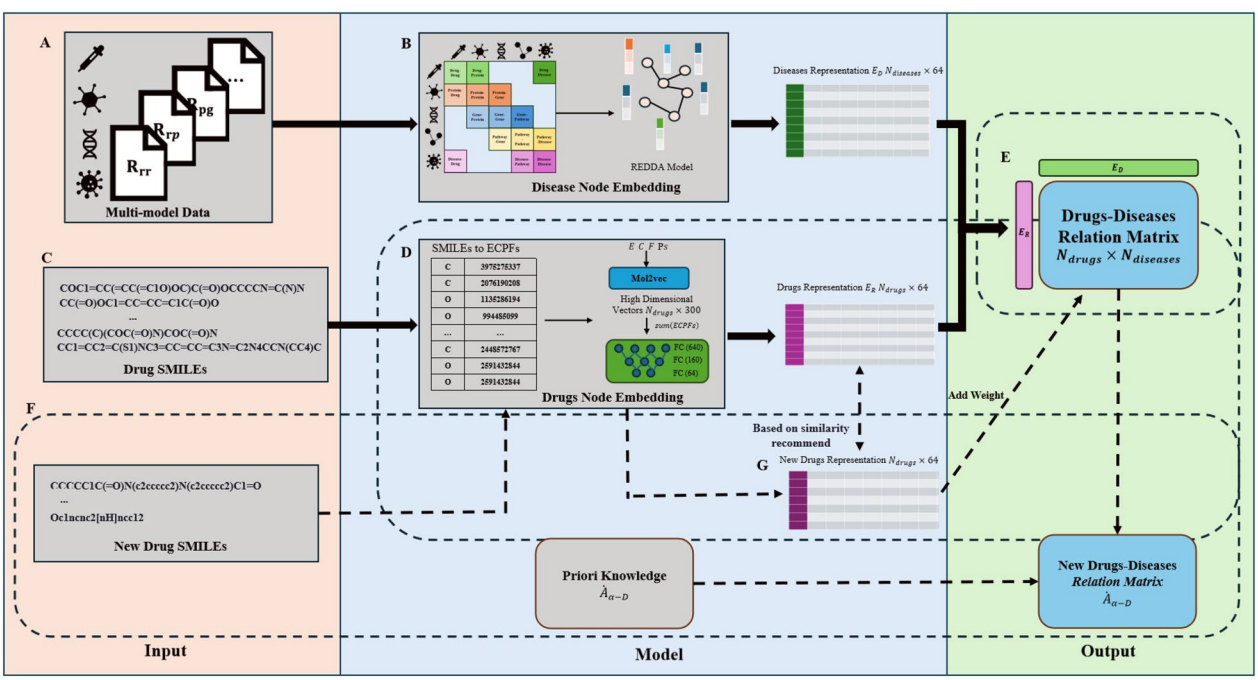

Fig. 1 SMRP flowchart. The solid box represents repositioning, and the dashed box represents the cold start. **A** Multimodal data. **B** The disease embedding module obtains disease embedding representations from heterogeneous networks through deep learning models. **C** The input of SMILE structural data. **D** The drug embedding module obtains a structure-based embedding representation. **E** Constructing a relationship matrix. **F** New drugs of the cold start module. **G** Weighting of the repositioning relationship matrix on the basis of feature similarity

Heterogeneous network construction and node feature initialization

First, a heterogeneous network G (N, E) is constructed on the basis of multimodal data as the input for the disease embedding module (Fig. 1A), where nodes include drugs, proteins, genes, pathways, and diseases. On the basis of the relationships organized in the dataset, a heterogeneous network is constructed, and the relationships are used as edge information in subsequent graph neural networks. Each node vector dimension is $N_{diseases} + N_{drugs}$. The features of drugs and diseases are assigned by their respective similarity matrices, and the remaining node features are initially set to zero. The SMILE structure data of the drug are organized as input for the drug embedding module (Fig. 1C).

Disease embedding module

The REDDA model is used as a disease embedding module [13]. This model is a complex graph neural network with an attention mechanism, which has received widespread attention for its excellent drug repositioning ability. The model first uses a fully connected network to initialize node features and map them to low dimensions. Then, two HeteroGCNs are used to learn the heterogeneous network globally, and one HeteroGCN subnetwork is used to learn each type of edge. Finally, one GAT module and one layer attention block are used to obtain a 64-dimensional embedded representation of diseases in the heterogeneous network, denoted as E_D.

Drug embedding module

To emphasize the consideration of the drug structure in the model and solve the subsequent cold start problem, the drug SMILE structure is used to generate embedded representations of drugs. First, SMILE, as a concise structural feature, includes molecular connectivity information. We use the ECFP to characterize the specific substructure of drug molecules [46]. Jaeger et al. [18], inspired by NLP, viewed the ECFP as a sentence and designed the MOL2VEC model to obtain embedded representations of substructures, improving the accuracy of molecular structure similarity retrieval. The model generates specific identifiers $ECFP_m$ for the substructures m around each heavy atom in SMILE and uses the unsupervised method MOL2VEC to convert $ECFP_m$ into a vector v_m. At the end of the module, the unbiased sum of all substructure vectors obtained from the ECFP in a molecule is used to obtain the representation V_R of the molecule. First, the formula that uses the MOL2VEC model to generate the vector representation of substructure m can be represented as:

$$v_m = Mol2Vec(ECFPm), m \in Substructures \tag{1}$$

Each drug embedding can be represented as:

$$V_R = \sum_{m \in Substructures} v_m \tag{2}$$

According to MOL2VEC's recommendation, a 300-dimensional feature vector can maximize the reflection of molecular structural information. To embed V_R, three layers of a fully connected network are used to encode the molecular representation and ultimately obtain a 64-dimensional drug embedding representation, denoted as E_R.

Building the drug–disease relationship matrix

For the embedding representations obtained from the disease embedding module and the drug embedding module, the model uses the product to obtain the drug–disease correlation matrix $\widehat{A}_{R-D}$ and normalizes it to a correlation score using the sigmoid function:

$$\widehat{A}_{R-D} = sigmoid\left(E_R {E_D}^T\right) \tag{3}$$

Drug cold start

To overcome the cold start problem, the SMPR model is inspired by the feature similarity-based recommendation system [41]. Our model's cold start is based on repositioning, providing users with a simple repositioning recommendation interface. When a new drug α does not appear in the dataset, the model only calls the drug embedding module. The user inputs the SMILE structure of the drug, and the model compares its embedding representation with the saved drug embedding representation database to recommend the most likely diseases to the user (Fig. 1F). The model first uses the reciprocal of the Euclidean distance to calculate the correlation ρ_α between E_α and E_R:

$$\rho_\alpha = \frac{1}{\sqrt{(E_{\alpha 1} - E_{R1})^2 + \ldots (E_{\alpha 64} - E_{R64})^2}} \tag{4}$$

The repositioning result $\widehat{A}_{R-D}$ is subsequently weighted using the correlation ρ_α, and the relationship between the new drug α and the existing drug $A_{\alpha-D}$ is used as prior knowledge. The formula for the relationship $\dot{A}_{\alpha-D}$ is:

$$\dot{A}_{\alpha-D} = \rho_\alpha \times \widehat{A}_{R-D} + A_{\alpha-D} \tag{5}$$

Then, $\dot{A}_{\alpha-D}$ is normalized:

$$\dot{A}_{\alpha-D} = \frac{A_{\alpha-R} - \min(A_{\alpha-R})}{\max(A_{\alpha-R}) - \min(A_{\alpha-R})} \tag{6}$$

Finally, diseases should be recommended for new drugs on the basis of the scores.

Model parameter and evaluation indicators

For the stability of the SMPR model results, we divided the dataset using tenfold cross validation and trained 4000 epochs at a time. The model removes the connection relationship of the test set in the graph during training. Since the repositioning task has a serious sample imbalance problem, the number of samples with a label of 0 is much larger than the number of samples with a label of 1; that is, the data contain fewer known drug–disease relationships. We chose the same approach as that employed in the existing studies and increase the model's attention to positive samples in the loss function. The model optimizer uses Adam, and the loss function uses a weighted BCEWithLogitsLoss function to balance the effects of different categories [58]. The formula can be denoted as:

$$Loss = -\frac{1}{N_D \times N_R}(\gamma \times \sum\nolimits_{(d,r)\in(N_{pos})} logPred_{d,r} + \sum\nolimits_{(d,r)\in(N_{neg})} \log(1 - Pred_{d,r})) \quad (7)$$

where N_D and N_R are the datasets of diseases and drugs, respectively. N_{neg} and N_{pos} are the number of relationships between labels 0 and 1 in the training set. (d, r) denotes the pair of disease and drug. γ is the weight parameter that is calculated by $\frac{N_{neg}}{N_{pos}}$, and $Pred_{d,r}$ is the predictive value of disease d and drug r. We chose the AUC, AUPR, F1 score, accuracy, recall, specificity, and precision proposed by Yu et al. [58] to evaluate our model. In the validation, error bars were added by calculating the standard deviation of each value in the cross validation. The AUC represents the area under the ROC curve, whereas the AUPR represents the area under the precision recall curve. The formulas can be denoted as follows:

$$AUC = \sum_{i=1}^{n-1}(FPR_{i+1} - FPR_i) \times \frac{TPR_i + TPR_{i+1}}{2} \quad (8)$$

where TPR is the true positive rate and where FPR is the false-positive rate.

$$AUPR = \sum_{i=1}^{n-1}(Recall_{i+1} - Recall_i) \times \frac{Precision_i + Precision_{i+1}}{2} \quad (9)$$

In addition, to evaluate the potential impact of structural features on model performance, we used the t-SNE algorithm [54] to reduce feature dimensionality as well as the K-Means unsupervised clustering algorithm [25]. We called the t-SNE function in the 1.2.2 version of the scikit-learn library to encode the 64-dimensional drug features into 2 dimensions for easy visualization analysis. Since the drug–disease relationship matrix R contains discrete data, for the evaluation score of the disease relationship between drug γ and drug δ, we use the following formula:

$$R_{\gamma-\delta} = R_{\gamma-D} + R_{\delta-D} \quad (10)$$

$$score = \frac{N_{\gamma-\delta_{i=2}}}{N_{\gamma-\delta_{i>1}}} \quad (11)$$

where $N_{\gamma-\delta_{i=2}}$ is the number of values equal to 2 and where $N_{\gamma-\delta_{i>1}}$ is the number of values greater than 1. Formula 10 adds the ground-truth disease relationships corresponding to drug γ and drug δ. In Formula 11, the number of added values of 2 is divided by the number of values greater than 1.

Results

Comparative experiment of drug repositioning models

We ran the repositioning task on DA and compared the results with those of the SMPR, REDDA, DRWBNCF [37], and DRAGNN [38] models. To ensure the fairness of the comparison results, tenfold cross validation was used for all the models. The performance results are shown in Table 3. After paying more attention to structural information, the prediction performance of the SMPR model indeed improved, especially in the distinction of positive samples, with a recall score of 69%. For the other evaluation indicators, the AUC was consistently better than that of the comparison model in the tenfold cross validation, reaching 0.98 (Fig. 2). The AUPR index reached 0.61, and the average value was the best in cross-validation. The F1 score and precision indicators of

Table 3 Model performance for DA

Method	AUC	AUPR	Accuracy	F1-score	Precision	Recall	Specificity
DRWBNCF	0.8774	0.2961	0.9914	0.3682	0.3591	0.3778	0.9955
DRAGNN	0.9217	0.5500	0.9949	0.5696	0.6524	0.5055	0.9982
REDDA	0.9755	0.4042	0.9897	0.3778	0.3164	0.4686	0.9932
SMPR	0.9870	0.6106	0.9934	0.5825	0.5015	0.6949	0.9954

the DRAGNN yielded the best results, while the SMPR scores were 0.58 and 0.50, respectively.

Sensitivity testing

We compared three main parameters in the sensitivity test of the model, including the hidden feature dimension, learning rate and patience. First, the dimension of the feature hidden layer contains the drug and disease feature information after encoding based on the heterogeneous graph network, which is used as the output of the disease embedding module and the drug embedding module. Three dimensions of *hidden_feats* ∈ {32,64,128,256} were selected for evaluation. Second, in the learning rate test, three values of *learningrate* ∈ {0.01,0.001,0.005} were used for testing. Finally, there is the patience parameter. Because the model contains an early stopping mechanism, the model evaluates whether the model has been fully learned by the F1 score indicator during training to prevent overfitting. Therefore, the patience indicator directly affects the number of times the model is trained. We compared the impacts of four values of *patience* ∈ {50,100,150,200} on the model results.

The sensitivity experiment is shown in Fig. 3. The results show that the hidden_feats parameter only has a significant drop in the output of 32 dimensions, and there is no significant difference in the other values. The model achieved the best result when the learning rate was 0.005. When the patience value was 50, the model stopped prematurely after approximately 600 training iterations, which had a significant impact on the results. When the patience value was 100 or above, the model results were similar, but the higher the patience value is, the more times the model will be trained, which is not conducive to preventing overfitting. On the basis of the results, we chose an output dimension of 64 dimensions, a learning rate of 0.005, and a patience of 100 as the parameters used in subsequent models.

Ablation experiments

The SMPR model comprises two main parts: the disease embedding module and the drug embedding module. The disease embedding module is composed of the REDDA model. In the design of the ablation experiment, we replaced it with a graph neural network. The model uses the GraphConv function provided by the DGL library of Python 3.10 to encode the heterogeneous network, and the parameters remain unchanged. Through the ReLU activation function and a layer of the fully connected network, it is embedded into a 64-dimensional output feature for the subsequent construction of the relationship network.

The other module is the drug embedding module, which uses the MOL2VEC model to encode the drug SMILE structure and embed it in the fully connected network. To verify the effectiveness of the MOL2VEC method in the drug repositioning task, we represented compound information in two ways. On one hand, we replaced it with the MACCS fingerprint. The MACCS fingerprint is also a commonly used compound representation method that can effectively represent molecular

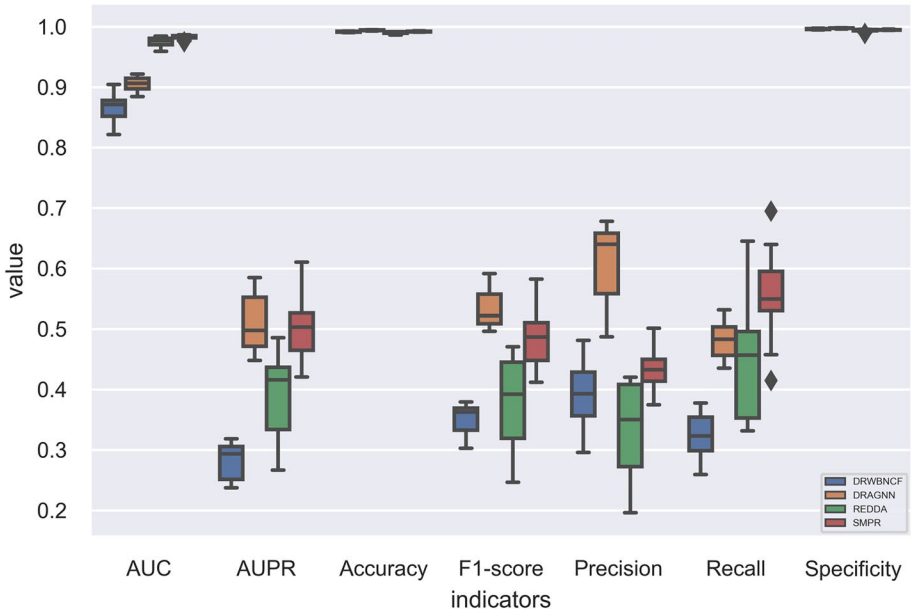

Fig. 2 Tenfold cross-validation results for the 4 models

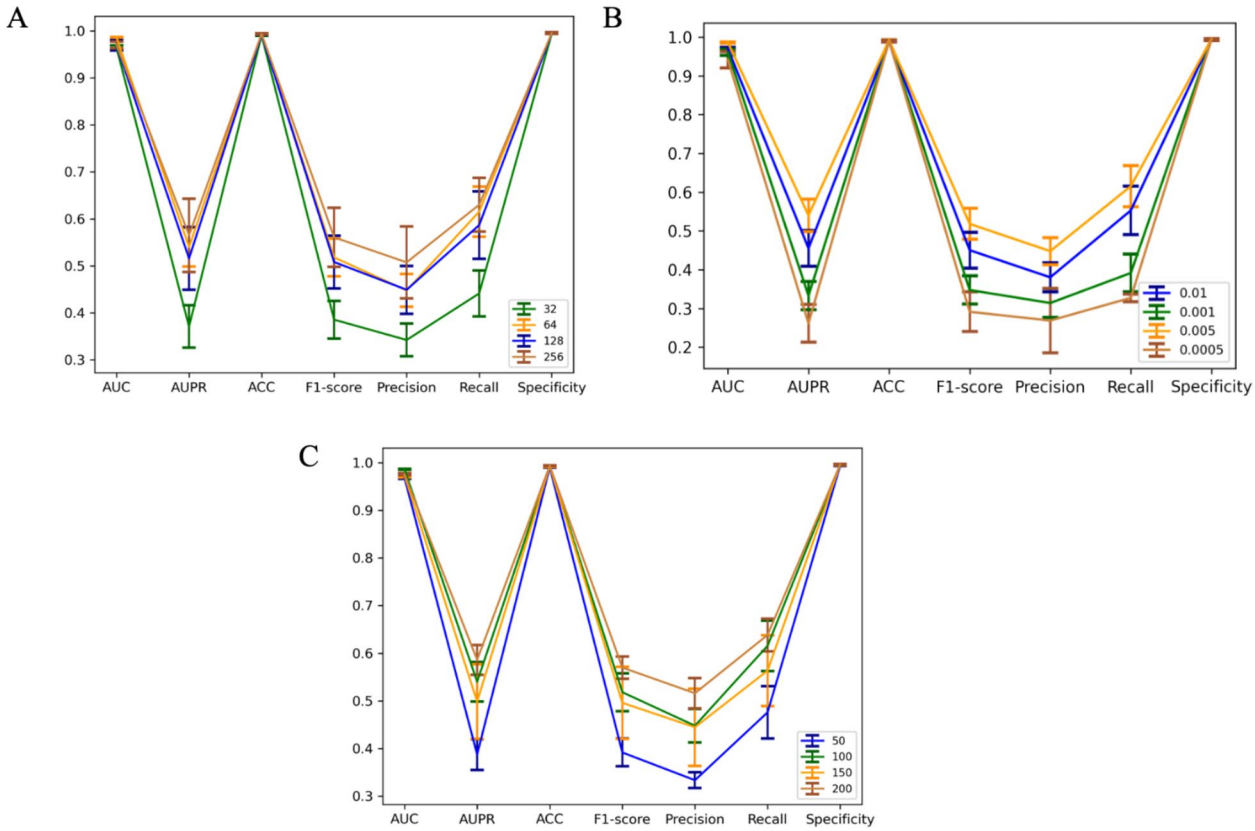

Fig. 3 Sensitivity experiment. The main parameters of the model include hidden features (**A**), the learning rate (**B**), and patience (**C**)

structure information. On the other hand, we encode compound structures through graph representations. Graph Isomorphism Network (GIN), an advanced graph neural network model, can directly encode the graph structure of compounds, effectively capturing molecular topology information and demonstrating strong performance in molecular representation learning [15]. DGL-LifeSci provides a standardized model that constructs a graph based on the input SMILE structure and encodes it using GIN's pre-trained model gin_supervised_infomax [30]. To maintain the rationality of the comparison, both representations retained 300-dimensional output features for the SMPR subsequent prediction module.

The results show that the SMPR model is significantly better than the model designed in the ablation experiment in terms of the AUC, F1 score, and precision. The AUPR, accuracy, and specificity indicators are comparable. However, the attenuation of the F1 score and precision indicators of the SMPR_GNN and SMPR_MACCS models is unacceptable, which shows that although they can better classify positive samples correctly, too many negative samples are also classified as positive. Using graph representation SMPR_GIN to encode drugs, AUC, accuracy, and precision indicators show good predictive performance, while recall indicator shows a significant downward trend slightly lower than SMPR. This also shows the effectiveness of the disease embedding module and drug embedding module as basic module designs for SMPR (Fig. 4).

Structural feature correlation analysis

We combined the t-SNE dimensionality reduction algorithm to reduce the structure-based features obtained by the drug structure embedding module to 2 dimensions and put them into the Cartesian coordinate system for observation. To determine the number of K-Means clusters, we combined the silhouette coefficient to evaluate the scores of clusters from 2 to 10 (Table 4). When the number of clusters was 6, the highest score reached 0.504. In Fig. 5A, it can be roughly judged that the data can be divided into 6 categories, and the K-Means unsupervised clustering method was used for clustering. The cluster boundaries in the figure are clear, and different categories can be clearly divided on the basis of structural features. Finally, we sorted the disease relationships in the corresponding dataset for each type of drug and visualized the results using a heatmap. In Fig. 5C, the disease similarity of the drugs is shown in a matrix, and the

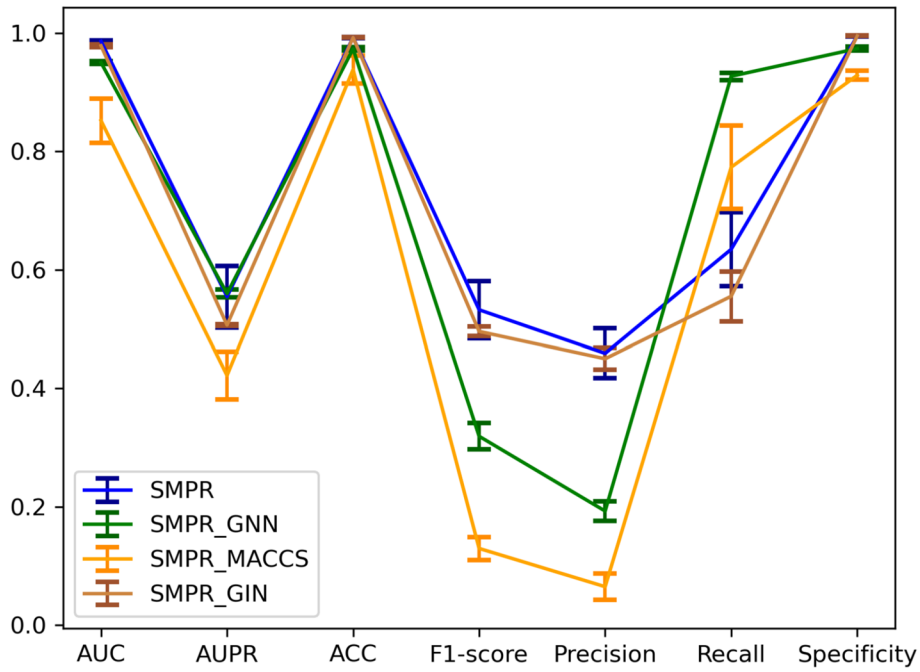

Fig. 4 Ablation experiments. SMPR is the original model. SMPR_GNN modifies the disease embedding module of the model into a GNN module. SMPR_MACCS replaced MOL2VEC in the drug embedding module, with the molecular structure represented by the MACCS fingerprint. SMPR_GIN replaced MOL2VEC with GIN

Table 4 K-means silhouette coefficient score

Clusters	2	3	4	5	6	7	8	9	10
Score	0.3725	0.4138	0.4242	0.4666	0.5043	0.4977	0.4829	0.4652	0.4505

red line distinguishes different categories. The value represents the average similarity between each category. On the diagonal of the matrix, the same type of drug has the highest similarity in the diseases on which they act. However, between different types, the similarity score of the disease they act on is significantly lower. This finding also proves that drugs with similar structures have similar binding pockets, which may lead to potential similarities in the diseases on which they act. The enhanced focus on structure in our model can effectively improve the prediction performance.

To further discuss the biological interpretability between different categories, we investigated the five most relevant diseases for each drug class (Table 5). Five out of the six types of drugs showed potential associations. Type 1 is classified as an abnormal response of the immune system. Type 2 comprises malignant tumors. Type 4 is an imbalance in the neurotransmitter system. Type 5 diseases are related mainly to metabolic and endocrine abnormalities. In Type 6, diseases related to vascular or endothelial cell damage are classified as progressive microvascular and stromal injury-related diseases. The only category with no obvious commonality is Type 3, but osteoarthritis, rheumatoid arthritis, and hyperlipidemia are associated with chronic inflammation.

The number of drugs in each category is indicated in parentheses in the category section.

Sparse matrix tests

The SMPR model uses a variety of entities to enrich the connection relationships of heterogeneous networks. We hope that the model can alleviate the adverse effects of many unknown relationships in this way. Therefore, we conducted sparse matrix tests in DA. As described in an existing study [51], the sparsity test removes approximately 20% of the drug–disease connections in the network. Here, to test the limits of our model's robustness, we removed $\varepsilon \in \{0\%, 20\%, 40\%, 60\%, 80\%, 100\%\}$ of the edges. The results are shown in Fig. 6. The AUPR index was stable at 20%, 40%, and 60% and was slightly lower than that of the complete dataset. When the percentage of deleted edges reached 80%, the AUC, AUPR and recall

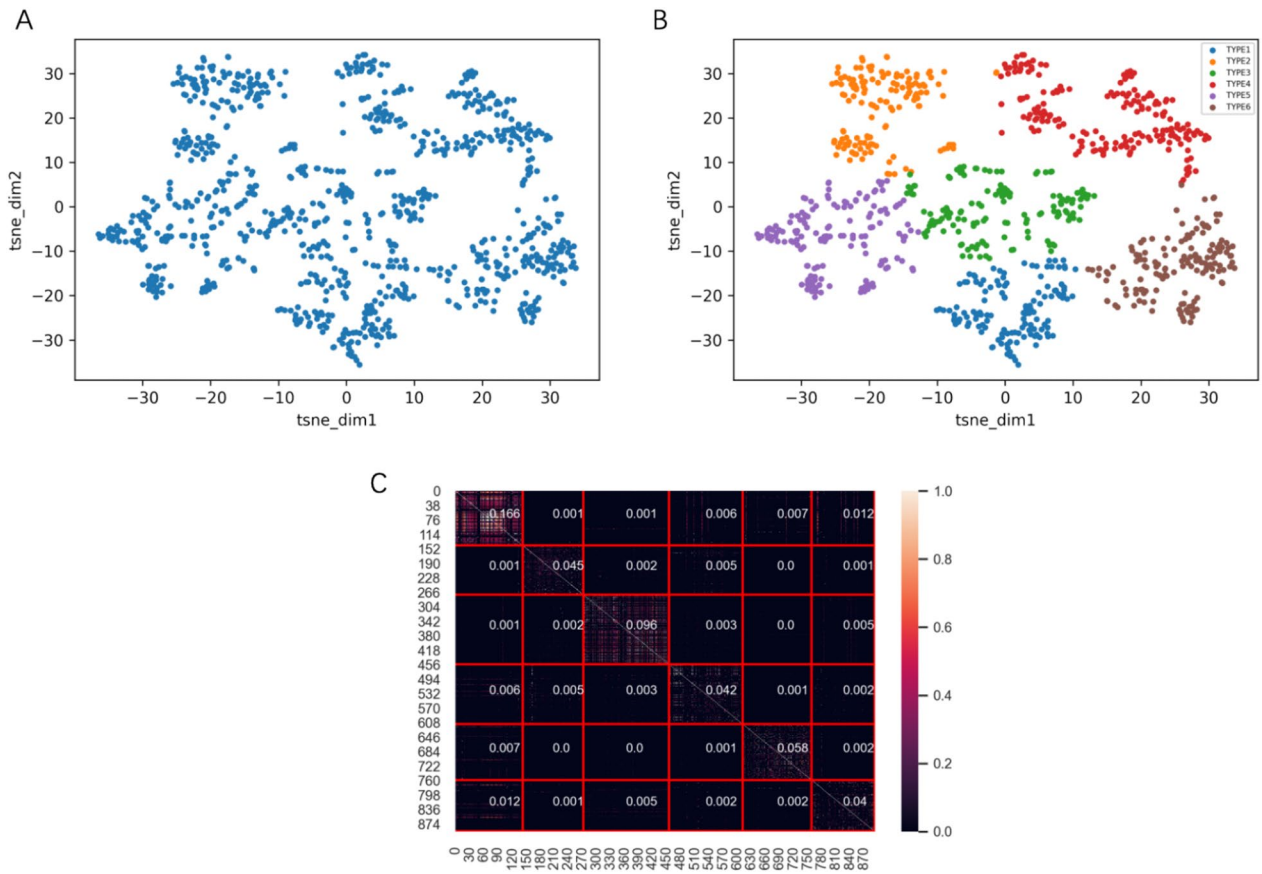

Fig. 5 Analysis of the potential effects of structural features. **A** Visualization results after t-SNE dimension reduction. **B** Combining the K-Means to distinguish classes. **C** Disease similarity scores of each drug–drug pair. The red line distinguishes different classes, and the value represents the average score of the comparison of different classes

scores were 0.968, 0.369, and 0.424, respectively, which are still within an acceptable range. When all drug–disease connections were deleted, the model indicators dropped significantly, and effective predictions could no longer be made. Therefore, for the 894 drugs, 454 diseases, and 2704 connections in DA, we needed only 540 drug–disease relationships for the model to be effectively trained, which means that our model has strong robustness.

Comparison results of DC

We also verified the evaluation indicators of the model in the DC to ensure the stability of the model in different public datasets. All four models passed the tenfold cross validation. The modal information contained in DC is limited and includes only drugs, proteins, and diseases. In addition, proteins do not have additional protein–protein interaction network (PPI) information. As shown in Fig. 7, the SMPR model still had the best prediction performance in terms of the AUC and AUPR. However, in terms of the F1 score and precision indicators, the DRAGNN model achieved the best results, whereas the REDDA and SMPR models performed poorly, with the same trend as for DA. We consider that the SMPR model judges more negative samples as positive, which may be a limitation of the model framework. Richer information can help improve the distinction between positive and negative samples. In addition, the SMPR model performs particularly well in terms of the recall indicator, showing the model's excellent predictive ability for positive samples.

Case study

We randomly selected two drugs from the SMPR repositioning results and conducted a literature review on the top 10 diseases that were most strongly associated with their predictions. Baclofen (DB00181) is a drug used to relieve severe muscle spasms caused by certain diseases of the brain and spinal cord. The top 10 recommended diseases are spasms and epilepsy. Most of the literature shows that baclofen has a therapeutic effect on target diseases, whereas some diseases indicate that

Table 5 The top 5 most relevant diseases for each category

	ID	Amount	Disease
TYPE 1 (131)	D009394	69	Nephritis, hereditary
	D006973	69	Hypertension
	D006319	16	Hearing loss, sensorineural
	D005902	14	Glaucoma, open-angle
	D001281	13	Atrial fibrillation
TYPE 2 (136)	D001932	31	Brain neoplasms
	D015451	21	Leukemia, lymphocytic
	D001943	20	breast neoplasms
	D015470	20	Leukemia, myeloid, acute
	D006689	20	Hodgkin disease
TYPE 3 (142)	D010003	29	Osteoarthritis
	D004401	19	Dysarthria
	D007592	19	Joint diseases
	D001172	13	Arthritis, rheumatoid
	D006949	13	Hyperlipidemias
TYPE 4 (184)	D001249	75	Respiratory hypersensitivity
	D003876	62	Dermatitis, atopic
	D009298	34	Nasal polyps
	D065631	31	Rhinitis, Allergic
	D011565	25	Psoriasis
TYPE 5 (156)	D003924	23	Diabetes mellitus, Type 2
	D001943	15	Breast neoplasms
	D005764	14	Gastroesophageal reflux
	D000152	11	Acne vulgaris
	D003922	11	Diabetes mellitus, Type 1
TYPE 6 (147)	D012559	29	Schizophrenia
	D010300	18	Parkinson disease
	D020734	17	Parkinsonian disorders
	D001764	16	Blepharospasm
	D016584	15	Panic disorder

baclofen can cause dysregulation of regulatory factors, leading to disease (Table 6). The relationship pathways between baclofen and the 10 diseases and their connections are shown in Fig. 8. The five pathways with the highest connectivity are hsa04921 (oxytocin signaling pathway), hsa04723 (retrograde endocannabinoid signaling), hsa04080 (neuroactive ligand–receptor interaction), hsa04010 (MAPK signaling pathway), and hsa05412 (arrhythmogenic right ventricular cardiomyopathy).

Another case, docetaxel (DB01248), is an anticancer drug used to treat breast cancer, lung cancer, prostate cancer, head cancer, neck cancer and other cancers. Among the top 10 diseases recommended by the model, 9 are related to cancer, and the literature shows that monotherapy or combination therapy can have therapeutic effects on the corresponding diseases (Table 7). In addition, cancer-related pathways have more complex regulatory mechanisms and richer connections. The pathways with the highest connectivity (Fig. 9) are hsa04144 (Ras signaling pathway), hsa05218 (melanoma), hsa04210 (MAPK signaling pathway), hsa05220 (transcriptional misregulation in cancer), and hsa04213 (longevity regulating pathway).

Results of the drug cold start application

Moreover, our model focuses on the application of drug cold start. For a drug that is not in the dataset and whose SMILE structure is known only, the aim is that the model can provide a simple interface. Therefore, we applied SMPR to simulate the cold start problem in CS_test. First, we retrained the model on CS_train and, to ensure the reliability of the model, we used fivefold cross validation. CS_test was subsequently used as a new dataset for cold start task testing.

According to the results shown in Table 8, when SMPR was retrained on the CS_train dataset, the results were stable, with the AUC hovering at approximately 0.97 and the AUPR remaining at approximately 0.45. Figure 10 shows the cold start performance of CS_test. SMPR predicted new drugs for all diseases of CS_test on the basis of similarity, and the model assumed that diseases with a score greater than 0.24 have potential effects on the basis of the evaluation metric mechanism. Moreover, the model has a certain predictive ability, especially in the division of positive samples, with the recall indicator above 0.7. For the division of negative samples, we believe that since the cold start is based on the repositioning results, some originally unknown relationships are predicted to be related, which has an impact on the evaluation of the results.

In addition, to verify the significance of the cold start model on the basis of the structural similarity of the drug embedding module, we visualized the results of fold1 (Fig. 11). First, on the basis of the training results of CS_train, we performed t-SNE dimensionality reduction on the vector R_{train} obtained by the drug embedding module and the R_{test} obtained by CS_test in the module and combined it with K-Means clustering. The results show that new drugs can indeed be classified into different categories on the basis of structural information. We also sorted the drug embedding features R_{train} and R_{test} according to their distances and displayed them in a heatmap. We collected the ground label of each new drug and the repositioning results of its highly similar drugs and calculated the label score between the new drugs of CS_test and the drugs of CS_train. We considered scores greater than 1.7 to be correct classifications. The results show that the overall label score between the new drug and the drug on the left in the heatmap is higher than that on the right, indicating that the higher the structural similarity is, the more obvious the correlation between the disease and the

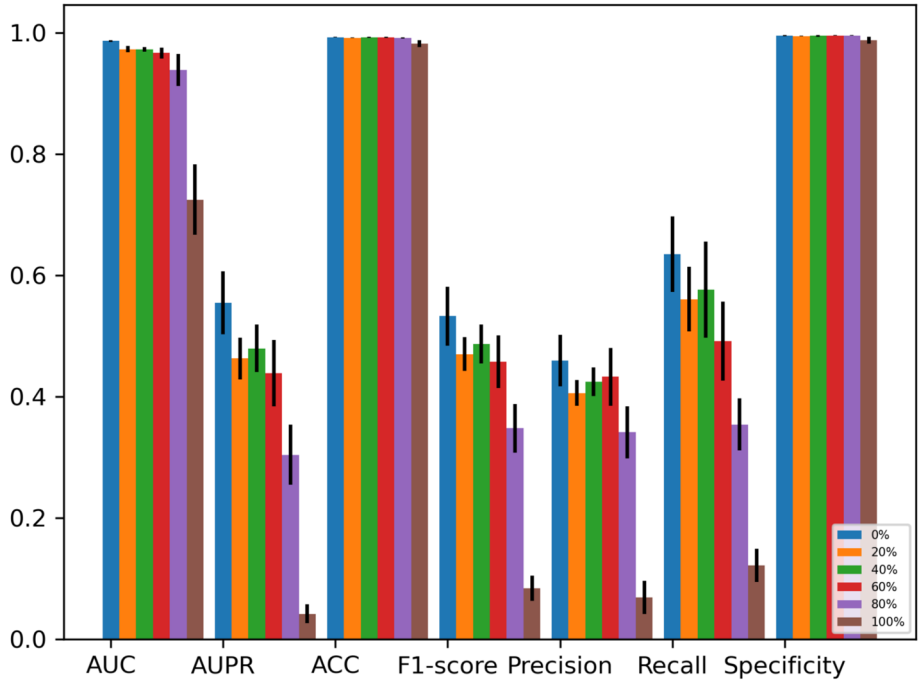

Fig. 6 The SMPR model was tested on sparse matrices in DA. The edges with $\varepsilon \in \{0\%, 20\%, 40\%, 60\%, 80\%, 100\%\}$ were deleted. The error bars are based on the standard deviation, with longer bars representing more discrete values

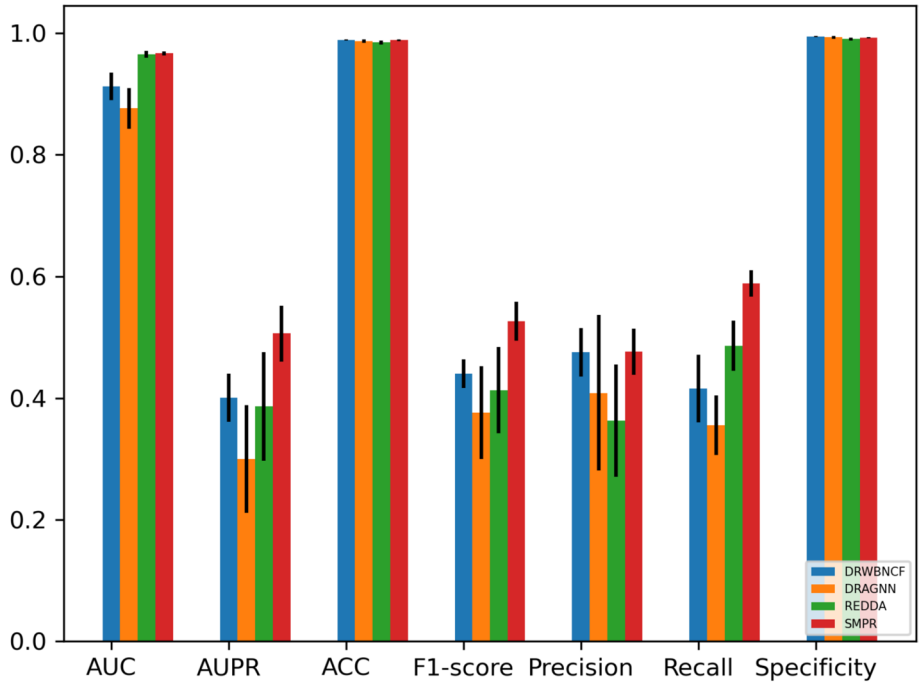

Fig. 7 Performance of the four models for DC

Table 6 Literature survey of DB00181 and the top 10 diseases

Drug	Disease	Description	Existing	Evidence
DB00181 Baclofen	D010003	Osteoarthritis	0	Abdelmonem et al. [1]
	D006816	Huntington disease	1	Shoulson et al. [49]
	D013132	Spinocerebellar degenerations	1	Bushart et al. [5]
	D009128	Muscle spasticity	1	Pérez-Arredondo et al. [44]
	D000690	Amyotrophic lateral sclerosis	1	Marquardt et al. [34]
	D015419	Spastic paraplegia, hereditary	1	Margetis et al. [33]
	D004401	Dysarthria	0	Leary et al. [27]
	D020190	Myoclonic epilepsy, juvenile	0	Akgun et al. [4]
	D004832	Epilepsy, absence	0	Inaba et al. [17]
	D004829	Epilepsy, generalized	0	O.Carter Snead [50]

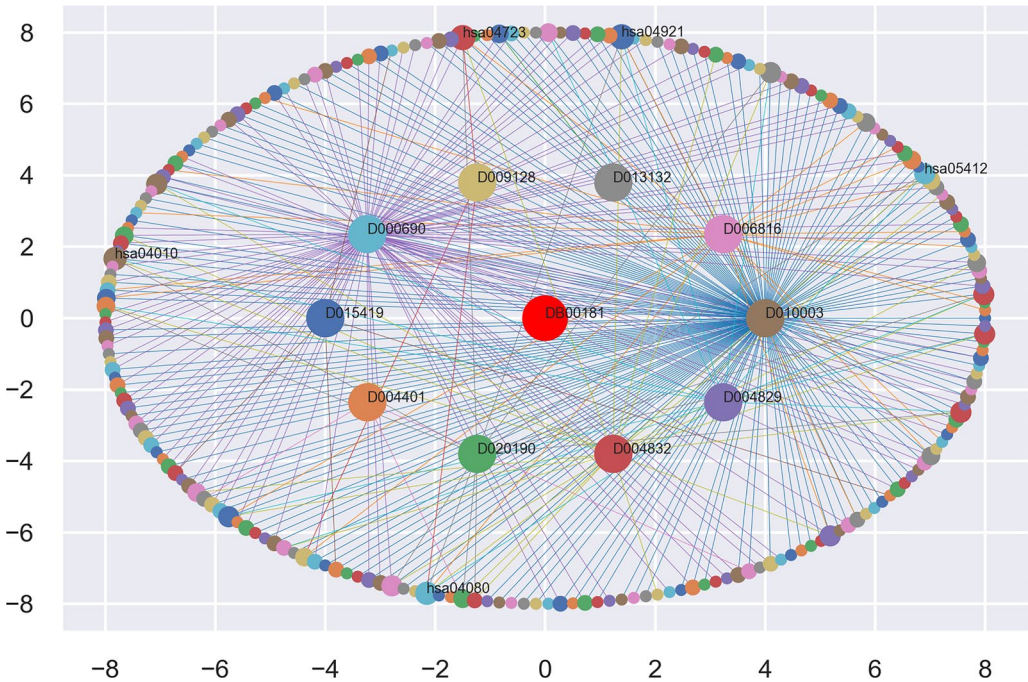

Fig. 8 Intersection of pathways between baclofen (DB00181) and the top 10 diseases. The top 5 pathways with the highest correlations are listed

drug. Therefore, the cold start model based on structural similarity in our task has a certain reference value.

SMPR local deployment

To reduce the difficulty of using the model further and improve its application potential, we deployed the model locally. Combined with the Qt programming tools, SMPR was packaged into a local executable file (.exe). An example of the software in operation is shown in Fig. 12. SMPR is a prediction model that is based on the results of repositioning deep learning; the packaged software provides the user with a SMILE structure input text box, and clicking the "Show Structure" button can display the input structure. When users click the 'Predict' button, the software generates a quick_predict.csv result locally on the basis of the input SMILE structure, which includes the scores of the input drug and 454 diseases.

Discussion

Traditional drug discovery work is long and arduous. Especially after experiencing the COVID-19 pandemic, we urgently need precise drug repositioning models to accelerate the process of drug discovery. However, existing models still have shortcomings in the application

Table 7 Literature survey of DB01248 and the top 10 diseases

Drug	Disease	Description	Existing	Evidence
DB01248 Docetaxel	D015470	Leukemia, myeloid, acute	0	Consolini et al. [7]
	D010190	Pancreatic neoplasms	0	Ryan et al. [47]
	D013274	Stomach neoplasms	1	Kazuhiro et al. [57]
	D001932	Brain neoplasms	1	Shaw et al. [48]
	D015179	Colorectal neoplasms	0	Wang et al. [55]
	D001749	Urinary bladder neoplasms	0	McKiernan et al. [36]
	D009447	Neuroblastoma	0	Francesco et al. [10]
	D011087	Polycythemia vera	0	Kunthur et al. [26]
	D002292	Carcinoma, renal cell	0	Marur et al. [35]
	D016889	Endometrial neoplasms	1	Miyahara et al. [39]

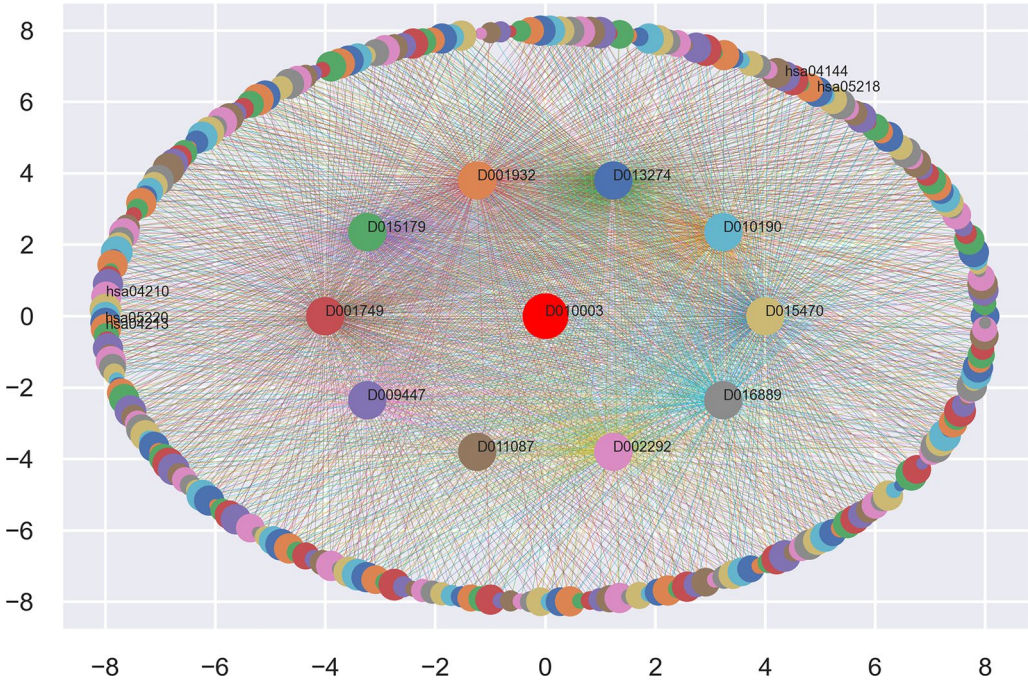

Fig. 9 Pathway intersection between docetaxel (DB01248) and the top 10 diseases. The pathways associated with cancer diseases are highly correlated

Table 8 Model performance for 5 folds in CS_train

	AUC	AUPR	Accuracy	F1-score	Precision	Recall	Specificity
fold1	0.9726	0.4967	0.9924	0.4866	0.4427	0.5402	0.9955
fold2	0.9709	0.4476	0.9913	0.4490	0.3875	0.5336	0.9944
fold3	0.9688	0.4409	0.9908	0.4377	0.3679	0.5402	0.9938
fold3	0.9739	0.4629	0.9909	0.4486	0.3751	0.5579	0.9938
fold5	0.9711	0.4535	0.9910	0.4535	0.3784	0.5658	0.9938

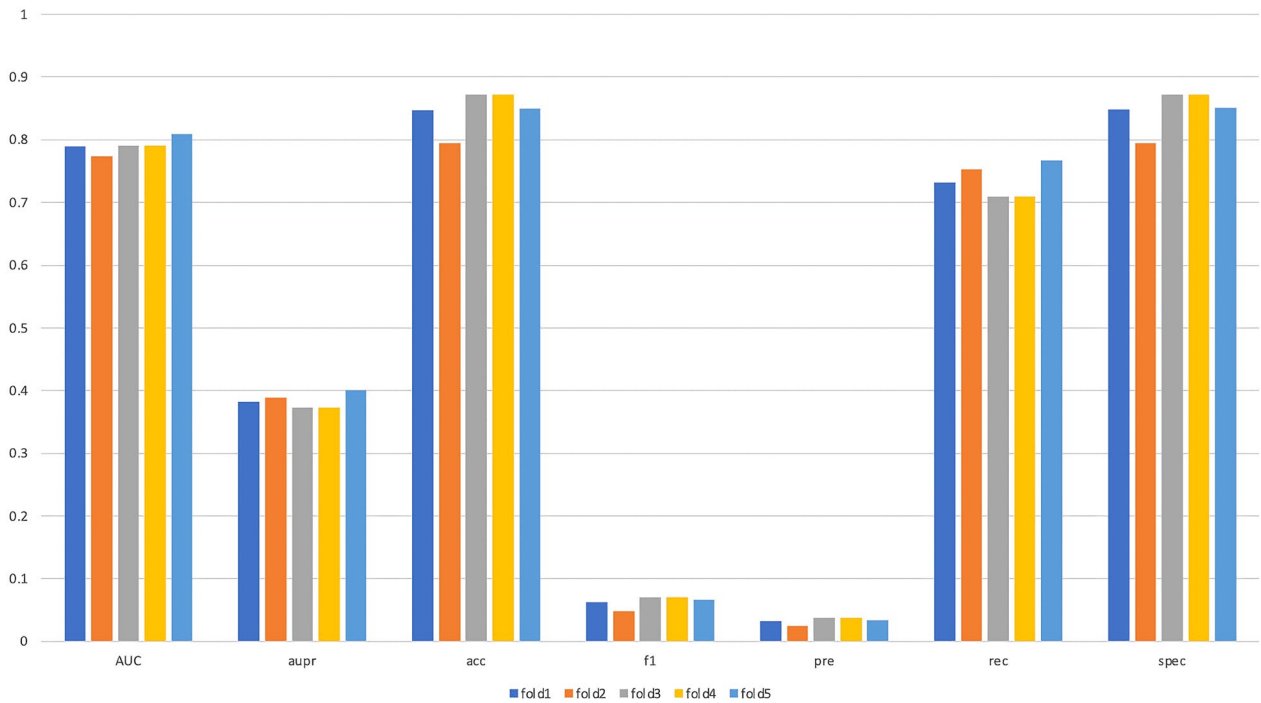

Fig. 10 SMPR cold start task performance for CS_test. Indicators include the AUC, AUPR, accuracy, F1 score, precision, recall and specificity

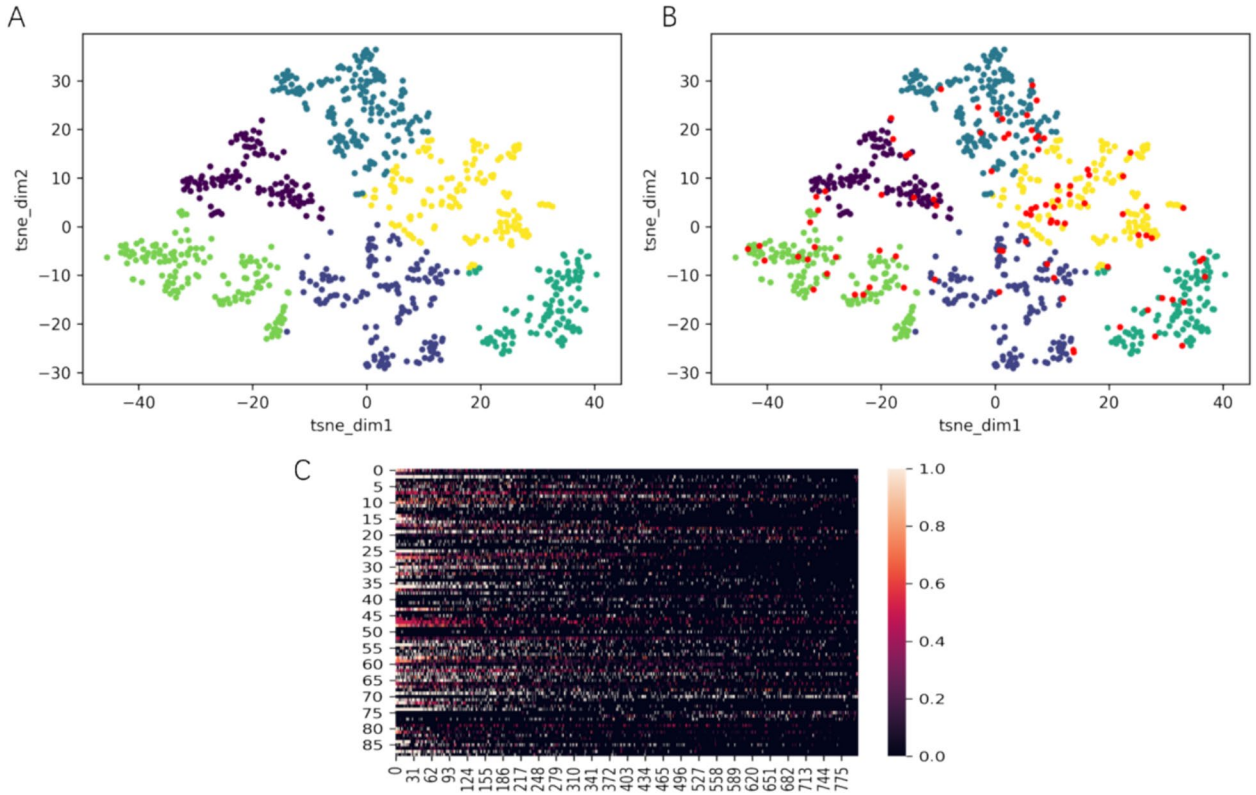

Fig. 11 Cold-start visualization analysis. **A** t-SNE and K-Means results of CS_train. **B** t-SNE result of CS_test. **C** The label scores sorted on the basis of drug embedding features R_{train} and R_{test} distance

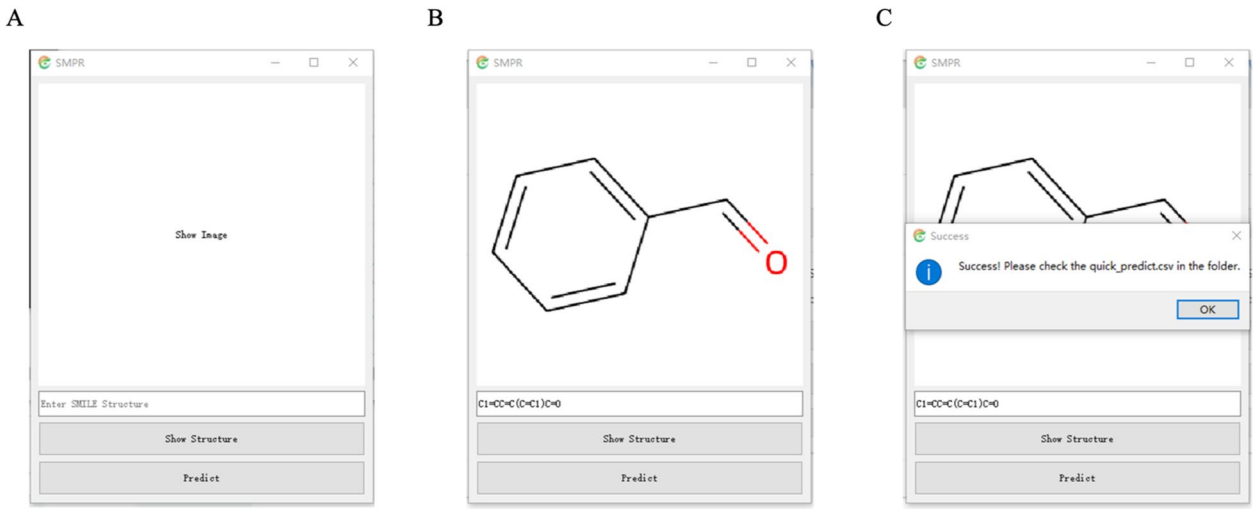

Fig. 12 SMPR local deployment. **A** The model is packaged into a quick-to-use executable program. **B** When the SMILE structure is entered and the 'Show Structure' button is clicked, the software will show the input drug. **C** When the 'Predict' button is clicked, a file of predicted results is generated locally

of structures. The search for potential new target proteins of drugs on the basis of structural similarity is also a major method for drug repositioning tasks. Li et al. [31] screened novel anti-inflammatory targets of nilotinib through molecular docking. The anticancer effect of benzimidazole was also discovered through a protein stoichiometry method [8]. These cases demonstrate the potential of structures in drug disease repositioning tasks. In addition, existing models also have poor generalizability. When facing the problem of a cold start, most models appear powerless, which makes it difficult to expand the knowledge of repositioning.

We propose a structure-enhanced multimodal prediction model, SMRP. This model has an enhanced focus on drug structure, and the results show that rich structural information has a beneficial effect on improving the accuracy of the model. Compared with existing models, our model achieves a performance AUC score of 98.7% and an AUPR score of 61.06%. Structural information has a favorable effect on the model's ability to predict drug–disease relationships, which is consistent with the fact that drugs with similar structures can enter the same binding pocket. In addition, in verification of the robustness of the model, SMPR yielded good prediction results in sparse matrix experiments and DC, which had less modal information. When at least 80% of the drug–disease relationships were retained, the model performance did not change significantly. Finally, a literature survey and analysis of two randomly selected drugs, baclofen (DB00181) and docetaxel (DB01248), and their effects on diseases also verified the application effect of the model, and multiple overlapping pathways were identified between diseases. Moreover, the model provides a cold start interface to facilitate the task of new drug prediction. Users need only enter the structural information of the drug, as well as optional supplementary information, to quickly make predictions. The visualization results show that different drugs can be well classified on the basis of structural information, and drugs with stronger structural correlation also have high correlation in their labels.

However, many parts of the model are still worth improving. With the rapid development of artificial intelligence technology, many new models are worth trying to implement [62]. The AlphaFold3 [2] model won the Nobel Prize in 2024, and researchers are increasingly focusing on the impact of structure. This also means that AlphaFold has greatly disclosed unknown protein structure information, which provides the possibility for subsequent models to introduce drug target binding stability and toxicity as auxiliary information. More detailed docking information will also increase the accuracy of the practical application of drug repositioning.

Conclusion

In summary, the SMPR model proposed in this study performs well in drug repositioning tasks and yielded stable prediction results on different datasets. In addition, the cold start module provided by the model can provide some reference when facing new drugs. This paper provides a simple executable program to reduce the difficulty of data processing for actual pharmacologists and accelerate the drug discovery process.

Abbreviations

SMPR	Structure-enhanced multimodal relationship prediction model
DA	Dataset A
ECFP	Extended connectivity fingerprint
MeSH	Medical subject headings
CS_train	Cold start training dataset
CS_test	Cold start testing dataset
DC	Dataset C
NLP	Natural language processing
PPI	Protein–protein interaction network
GIN	Graph isomorphism network

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Author contributions
X D conceived the work. X D and R M wrote the main manuscript text. SY Z designed the software. SB J and LF Z design figures. Y L, JH Z and YZ Z revised the manuscript. The authors read and approved the final manuscript.

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Availability of data and materials
The code, datasets and software of SMPR are provided at https://github.com/dxxxin/SMPR. The datasets supporting the conclusions of this article are available in the GitHub repository.

Declarations

Competing interests
The authors declare no competing interests.

Author details
[1]Faculty of Innovation Engineering, Macau University of Science and Technology, Avenida Wai Long, Taipa, Macao 999078, China. [2]School of Pharmacy, Macau University of Science and Technology, Avenida Wai Long, Taipa, Macao 999078, China. [3]Basic Teaching Department, Zhuhai Campus of Zunyi Medical University, Zhu Hai 519000, China. [4]Faculty of Humanities and Arts, Macau University of Science and Technology, Avenida Wai Long, Taipa, Macao 999078, China. [5]Medical Engineering Technology and Data Mining Institute, Zhengzhou University, 100 Science Avenue, Zhengzhou 450000, China. [6]Chinese Medicine Guangdong Laboratory (Hengqin), 2522 Huandao North Road, Zhu Hai 519031, China.

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